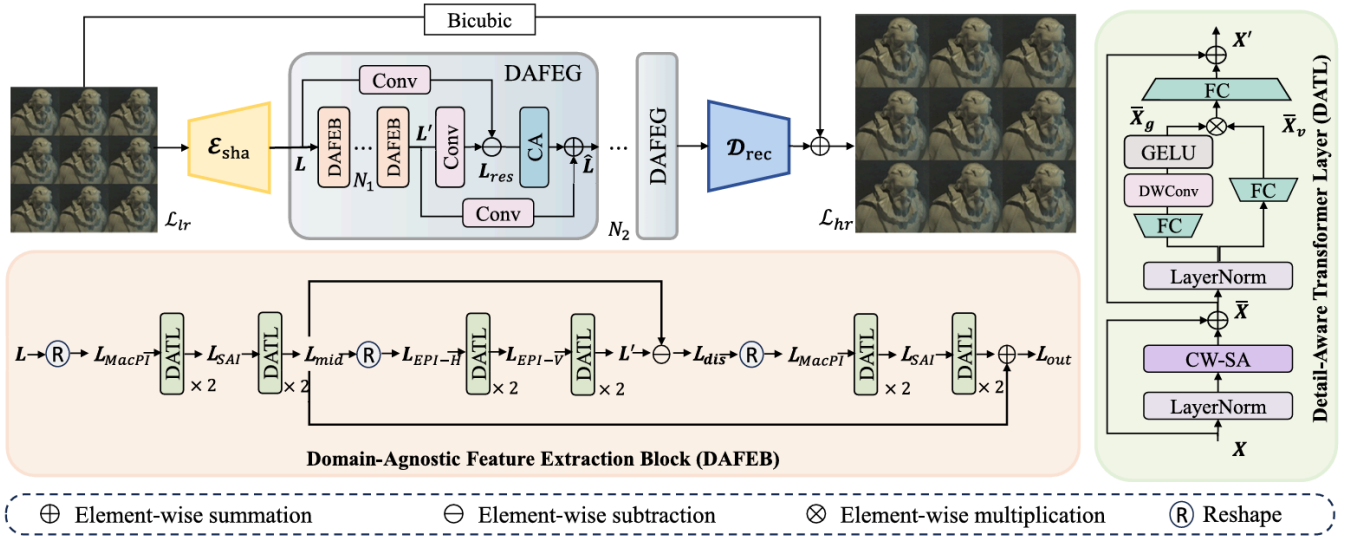


Learning Domain-Agnostic Spatial-Angular Feature for Light Field Image Super-Resolution

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Preparation:

1. Requirement:

- pytorch = 1.12.1, torchvision = 0.13.1, python = 3.8

2. Datasets:

- We use five LF benchmarks in [BasicLFSR](#) (i.e., EPFL, HCInew, HCIold, INRIA, and STFgantry). Download and put them in folder `./datasets/`.

3. Generate training and testing data:

- Run `Generate_Data_for_Training.py` to generate training data in `./data_for_training/`.
- Run `Generate_Data_for_Test.py` to generate testing data in `./data_for_test/`.

Train:

- Set the hyper-parameters in `parse_args()` in `train.py` if needed.
- Run `train.py` to train network
- Checkpoints will be saved in `./log/`

Test:

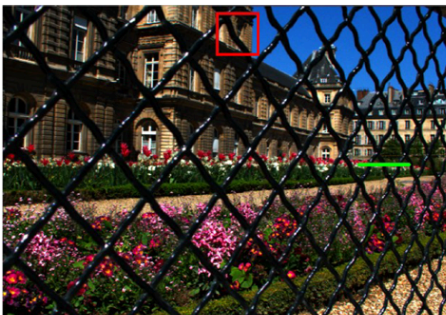
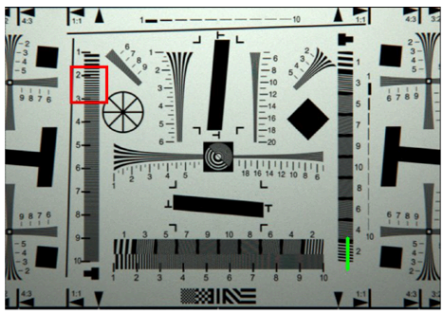
- Run `test.py` to perform test on each dataset. The resultant `.mat` files will be saved in `./Results/`
- Run `GenerateResultImages.py` to generate SR RGB images. Saved in `./SRimage/`

Results:

Quantitative results:

Method	Scale	#Param.(M)	FLOPs(G)	EPFL	HCInew	HCold	INRIA	STFgantry	Average
Bicubic	x2	–	–	29.50/.9350/.164	31.69/.9335/.198	37.46/.9776/.106	31.10/.9563/.177	30.82/.9473/.127	31.11/.9542/.154
RCAN [8]	x2	15.3	388.69	33.16/.9635/.111	34.98/.9602/.122	41.05/.9875/.051	35.01/.9769/.138	36.33/.9825/.033	36.10/.9741/.091
SwinIR [9]	x2	11.70	300.28	31.81/.9213/.110	34.14/.9175/.138	40.10/.9713/.055	32.94/.9061/.139	36.94/.9799/.039	35.18/.9395/.096
HAT [52]	x2	40.17	1034.61	34.89/.9478/.108	36.44/.9324/.132	42.59/.9792/.050	37.26/.9578/.137	39.28/.9854/.032	38.09/.9605/.091
resLF [12]	x2	7.98	37.06	32.75/.9672/.094	36.07/.9715/.078	42.61/.9922/.028	34.57/.9784/.128	36.89/.9873/.028	36.58/.9793/.071
LFSSR [10]	x2	0.88	25.70	33.69/.9748/.086	36.86/.9753/.078	43.75/.9939/.026	35.27/.9834/.122	38.07/.9902/.031	37.73/.9835/.068
LF-InterNet [15]	x2	5.04	47.46	34.14/.9761/.083	37.28/.9769/.074	44.45/.9945/.023	35.80/.9846/.119	38.72/.9916/.028	38.08/.9847/.065
LF-ATO [13]	x2	1.22	597.66	34.27/.9757/.084	37.24/.9767/.071	44.20/.9942/.024	36.15/.9842/.119	39.64/.9929/.021	38.15/.9843/.064
MEG-Net [14]	x2	1.69	48.40	34.34/.9773/.078	37.42/.9777/.067	44.08/.9942/.023	36.09/.9849/.114	38.77/.9915/.026	38.14/.9851/.062
LF-DFNet [16]	x2	3.94	57.22	34.44/.9766/.082	37.44/.9786/.072	44.23/.9943/.022	36.36/.9841/.118	39.61/.9935/.017	38.41/.9854/.062
IINet [31]	x2	4.84	56.16	34.68/.9773/.078	37.74/.9790/.062	44.84/.9948/.021	36.57/.9853/.115	39.86/.9936/.020	38.74/.9857/.059
LF-SAV [19]	x2	1.22	34.65	34.62/.9772/.081	37.43/.9776/.073	44.22/.9942/.026	36.36/.9849/.120	38.69/.9914/.027	38.26/.9851/.065
DistgSSR [17]	x2	3.53	64.11	34.81/.9787/.075	37.96/.9796/.061	44.94/.9949/.020	36.59/.9859/.011	40.40/.9942/.017	38.94/.9866/.057
HLFSR [18]	x2	13.72	167.40	<u>35.31/.9800/.070</u>	<u>38.32/.9807/.057</u>	44.98/.9950/.020	<u>37.06/.9867/.106</u>	40.85/.9947/.016	39.30/.9874/.054
DPT [20]	x2	3.73	65.34	34.48/.9758/.083	37.35/.9771/.070	44.31/.9943/.024	36.40/.9843/.118	39.52/.9926/.022	38.40/.9848/.063
LFT [21]	x2	1.11	56.16	34.80/.9781/.075	37.84/.9791/.063	44.52/.9945/.022	36.59/.9855/.111	40.51/.9941/.017	38.85/.9863/.058
EPIT [22]	x2	1.42	69.71	34.83/.9775/.080	38.23/.9810/.056	<u>45.08/.9949/.021</u>	36.67/.9853/.115	42.17/.9957/.012	39.40/.9877/.057
LF-DET [23]	x2	1.59	48.50	35.26/.9797/.071	38.31/.9807/.057	44.99/.9950/.020	36.95/.9864/.106	<u>41.76/.9955/.013</u>	<u>39.45/.9875/.053</u>
LF-DANet (Ours)	x2	1.30	48.08	36.30/.9839/.060	38.49/.9814/.055	45.32/.9952/.020	37.71/.9883/.096	41.54/.9954/.015	39.90/.9888/.049
Bicubic	x4	–	–	25.14/.8311/.362	27.61/.8507/.405	32.42/.9335/.302	26.82/.8860/.384	25.93/.8431/.303	27.58/.8701/.351
RCAN [8]	x4	15.4	390.12	27.88/.8863/.296	29.63/.8880/.326	35.20/.9540/.224	29.76/.9273/.315	28.90/.9110/.177	30.36/.9141/.268
SwinIR [9]	x4	11.80	315.91	27.12/.8052/.274	28.64/.7853/.326	34.22/.8933/.212	28.50/.8036/.297	29.50/.9048/.155	29.59/.8384/.252
HAT [52]	x4	40.32	1050.24	29.72/.8531/.260	30.92/.8187/.315	36.66/.9169/.202	31.89/.8908/.286	31.56/.9231/.135	32.15/.8805/.239
resLF [12]	x4	8.64	39.70	28.27/.9035/.271	30.73/.9107/.275	36.71/.9682/.166	30.34/.9412/.297	30.19/.9372/.155	31.25/.9322/.232
LFSSR [10]	x4	1.77	128.44	28.27/.9118/.260	30.72/.9145/.270	36.70/.9696/.162	30.31/.9467/.286	30.15/.9426/.147	31.23/.9370/.225
LF-InterNet [15]	x4	5.48	50.10	28.67/.9162/.246	30.98/.9161/.263	37.11/.9716/.146	30.64/.9491/.277	30.53/.9409/.154	31.58/.9388/.217
LF-ATO [13]	x4	1.36	686.99	28.52/.9115/.261	30.88/.9135/.272	37.00/.9699/.160	30.71/.9484/.283	30.61/.9430/.143	31.54/.9373/.223
MEG-Net [14]	x4	1.77	102.20	28.74/.9160/.249	31.10/.9177/.258	37.28/.9716/.147	30.66/.9490/.279	30.77/.9453/.142	31.71/.9399/.215
LF-DFNet [16]	x4	3.99	57.31	28.77/.9165/.241	31.23/.9196/.262	37.32/.9718/.142	30.83/.9503/.270	31.15/.9494/.116	31.86/.9415/.206
IINet [31]	x4	4.88	57.42	29.11/.9188/.244	31.36/.9208/.248	37.62/.9734/.136	31.08/.9515/.273	31.21/.9502/.124	32.08/.9429/.205
LF-SAV [19]	x4	1.54	115.80	29.37/.9223/.244	31.45/.9217/.251	37.50/.9721/.150	31.27/.9531/.273	31.36/.9505/.128	32.19/.9439/.209
DistgSSR [17]	x4	3.58	65.41	28.99/.9195/.244	31.38/.9217/.246	37.56/.9732/.132	30.99/.9519/.271	31.65/.9535/.117	32.11/.9440/.202
HLFSR [18]	x4	13.87	182.52	29.20/.9222/.236	<u>31.57/.9238/.239</u>	37.78/.9742/.129	31.24/.9534/.266	31.64/.9537/.115	32.29/.9455/.196
DPT [20]	x4	3.78	66.55	28.93/.9170/.248	31.19/.9188/.256	37.39/.9721/.145	30.96/.9503/.275	31.14/.9488/.128	31.92/.9414/.210
LFT [21]	x4	1.16	57.60	29.25/.9210/.240	31.46/.9218/.247	37.63/.9735/.135	31.20/.9524/.271	31.86/.9548/.117	32.28/.9447/.202
EPIT [22]	x4	1.47	74.96	29.34/.9197/.245	31.51/.9231/.244	37.68/.9737/.136	31.37/.9526/.271	<u>32.18/.9571/.109</u>	32.40/.9452/.201
LF-DET [23]	x4	1.69	51.20	<u>29.47/.9230/.236</u>	31.56/.9235/.240	<u>37.84/.9744/.132</u>	<u>31.39/.9534/.270</u>	<u>32.14/.9573/.107</u>	<u>32.48/.9463/.197</u>
LF-DANet (Ours)	x4	1.35	49.35	30.02/.9280/.227	31.84/.9268/.231	38.09/.9755/.128	31.98/.9565/.261	32.71/.9609/.100	32.93/.9495/.189

Visual comparisons:

 INRIA_Sculpture × 2	HR	resLF	LFSSR	LF-ATO	LF-InterNet	IINet	MEG-Net	LF-DFNet
	PSNR/SSIM	34.32/0.9816	34.12/0.9809	35.13/0.9821	34.83/0.9826	35.88/0.9839	35.02/0.9832	35.61/0.9832
	LF-SAV	DistgSSR	HLFSR	DPT	LFT	EPIT	LF-DET	Ours
	35.15/0.9833	35.82/0.9847	36.43/0.9857	35.73/0.9826	35.61/0.9841	35.86/0.9829	36.44/0.9852	36.80/0.9876
 EPFL_Palais_du_Luxembourg × 2	HR	resLF	LFSSR	LF-ATO	LF-InterNet	IINet	MEG-Net	LF-DFNet
	PSNR/SSIM	30.49/0.9605	30.45/0.9601	30.54/0.9605	30.71/0.9628	31.02/0.9653	30.92/0.9647	30.70/0.9618
	LF-SAV	DistgSSR	HLFSR	DPT	LFT	EPIT	LF-DET	Ours
	30.94/0.9648	31.21/0.9667	31.37/0.9683	30.65/0.9619	31.17/0.9665	30.83/0.9632	31.46/0.9688	32.91/0.9770
 EPFL_ISO_Chart_1 × 4	HR	resLF	LFSSR	LF-ATO	LF-InterNet	IINet	MEG-Net	LF-DFNet
	PSNR/SSIM	25.95/0.9272	26.25/0.9254	26.16/0.9338	26.54/0.9320	26.83/0.9372	26.82/0.9333	28.15/0.9442
	LF-SAV	DistgSSR	HLFSR	DPT	LFT	EPIT	LF-DET	Ours
	27.68/0.9444	27.22/0.9407	27.52/0.9449	26.81/0.9363	27.37/0.9414	27.87/0.9437	27.65/0.9413	29.10/0.9519
 STFgantry_Tarot_Cards_S × 4	HR	resLF	LFSSR	LF-ATO	LF-InterNet	IINet	MEG-Net	LF-DFNet
	PSNR/SSIM	27.62/0.9163	27.76/0.9182	27.70/0.9188	27.69/0.9181	28.12/0.9260	27.99/0.9225	28.22/0.9273
	LF-SAV	DistgSSR	HLFSR	DPT	LFT	EPIT	LF-DET	Ours
	28.19/0.9259	28.43/0.9302	28.47/0.9306	27.95/0.9220	28.60/0.9319	28.63/0.9326	28.75/0.9346	29.08/0.9384

Citation

If you find LF-DANet useful in your research or projects, please cite our work:

```
1 @article{xia2026learning,  
2   title={Learning Domain-Agnostic Spatial-Angular Feature for Light Field Image Super-  
   Resolution},  
3   author={Xia, Wang and Lu, Yao and Wang, Shunzhou and Xu, Lingwen},  
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